EXPERIMENT-5

DATA VISUALIZATION WITH MATPLOTLIB

AIM

To visualize and analyze the given dataset using various plots such as line charts, bar graphs, pie charts, and histograms with the help of the Matplotlib library, for better interpretation and understanding of trends and patterns.

PREREQUISITES & REQUIREMENTS:

1. Computer with Python Installed
2. A dataset for Practice
3. Jupytor Notebook
4. Knowledge on Pandas & Matplotlib Libraries

MATPLOTLIB

1. Matplotlib is a popular python library for Data Visualization.
2. It can be used to draw 2-D plots from data and arrays.
3. It supports almost every plot types such as line plots, scatter plots, bar charts, histograms etc.
4. Many other visualization libraries such as seaborn, python built-in visualizations are directly built with Matplotlib. \
5. Matlotlib is inspired by the plotting functions of MatLab programming language.
6. In Matplotlib, creating plots can be done with functional based methods and OOP based methods.
7. For more useful information about matplotlib - https://matplotlib.org/2.0.2/gallery.html

Step 1: Import the required libraries & dataset

In [8]:
```
import pandas as pd
import matplotlib.pyplot as plt

# Read the renewable energy dataset
df_241UDS0096 = pd.read_csv("renewable_energy_dataset.csv")

df_241UDS0096.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 16 entries, 0 to 15
Data columns (total 8 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   Year                         16 non-null     int64
 1   Solar_Capacity_GW            16 non-null     float64
 2   Wind_Capacity_GW             16 non-null     float64
 3   Hydro_Capacity_GW            16 non-null     float64
 4   Renewable_Generation_TWh     16 non-null     float64
 5   Investment_Crore             16 non-null     int64
 6   Energy_Demand_TWh            16 non-null     float64
 7   Total_Renewable_Capacity_GW  16 non-null     float64
dtypes: float64(6), int64(2)
memory usage: 1.1 KB
```

Step 2: Display the dataset

In [9]:
```
df_241UDS0096.head()
```

Out[9]:

	Year	Solar_Capacity_GW	Wind_Capacity_GW	Hydro_Capacity_GW	Renewable_Generation_TWh	Investment_Crore	Energy_Demand_TWh	Total_Renewable_Cap
0	2010	3.84	11.88	37.99	92.41	20844	844.51	53.71
1	2011	9.41	16.07	38.29	92.33	29880	918.93	63.77
2	2012	18.54	17.20	40.92	121.60	32015	1006.95	76.66
3	2013	27.89	19.19	40.42	131.30	42912	1017.04	87.50
4	2014	30.65	25.23	42.70	143.93	47799	1069.79	98.58

In [10]:
```
df_241UDS0096
```

Out[10]:

	Year	Solar_Capacity_GW	Wind_Capacity_GW	Hydro_Capacity_GW	Renewable_Generation_TWh	Investment_Crore	Energy_Demand_TWh	Total_Renewable_Ca
0	2010	3.84	11.88	37.99	92.41	20844	844.51	53.71
1	2011	9.41	16.07	38.29	92.33	29880	918.93	63.77
2	2012	18.54	17.20	40.92	121.60	32015	1006.95	76.66
3	2013	27.89	19.19	40.42	131.30	42912	1017.04	87.50
4	2014	30.65	25.23	42.70	143.93	47799	1069.79	98.58
5	2015	37.81	25.80	42.10	167.62	51409	1137.46	105.71
6	2016	49.51	28.74	43.74	185.22	62065	1232.89	121.99
7	2017	54.64	29.54	46.09	199.19	73316	1278.22	130.27
8	2018	58.71	33.19	47.66	201.46	74941	1316.76	139.56
9	2019	68.40	36.57	48.34	219.84	87676	1402.83	153.31
10	2020	73.04	37.65	49.24	238.99	80164	1452.43	159.93
11	2021	80.20	42.08	50.23	258.16	99343	1534.22	172.51
12	2022	89.12	43.50	50.42	262.65	103905	1552.45	183.04
13	2023	90.90	46.46	52.16	279.37	109687	1621.81	189.52
14	2024	98.53	48.68	53.50	287.59	118188	1680.20	200.71
15	2025	108.59	54.22	55.85	301.63	118044	1713.41	218.66

In [11]:
```
df_241UDS0096.columns
```

Out[11]:
```
Index(['Year', 'Solar_Capacity_GW', 'Wind_Capacity_GW', 'Hydro_Capacity_GW',
       'Renewable_Generation_TWh', 'Investment_Crore', 'Energy_Demand_TWh',
       'Total_Renewable_Capacity_GW'],
      dtype='object')
```

In [13]:
```
#Convert Year into a suitable format
df_241UDS0096['Year'] = pd.to_numeric(df_241UDS0096['Year'])
```

Step 3: Visualize Data Using Various Plots

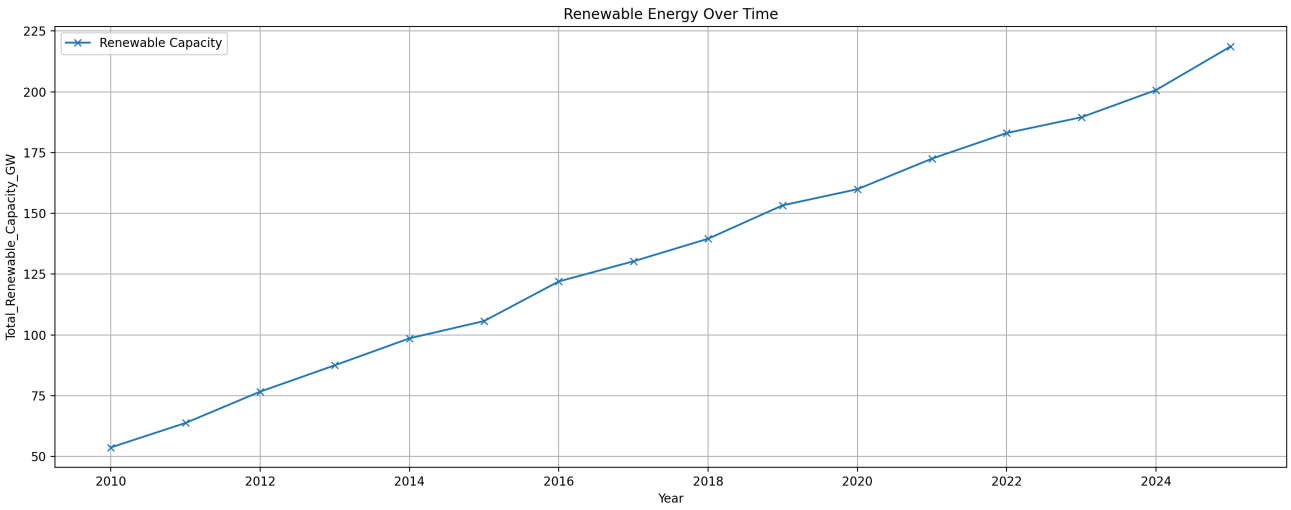
In [19]:
```
#1. Renewable Energy Over Time
plt.figure(figsize=(15, 6), dpi=200)

plt.plot(
    df_241UDS0096['Year'],
    df_241UDS0096['Total_Renewable_Capacity_GW'],
    marker='x',
    label='Renewable Capacity'
)

plt.title('Renewable Energy Over Time')
plt.xlabel('Year')
plt.ylabel('Total_Renewable_Capacity_GW')

plt.tight_layout()
plt.legend()
plt.grid(True)

plt.show()
```



Step 2: Subplots

In [24]:
```
fig, axes = plt.subplots(1, 2, figsize=(20, 4))

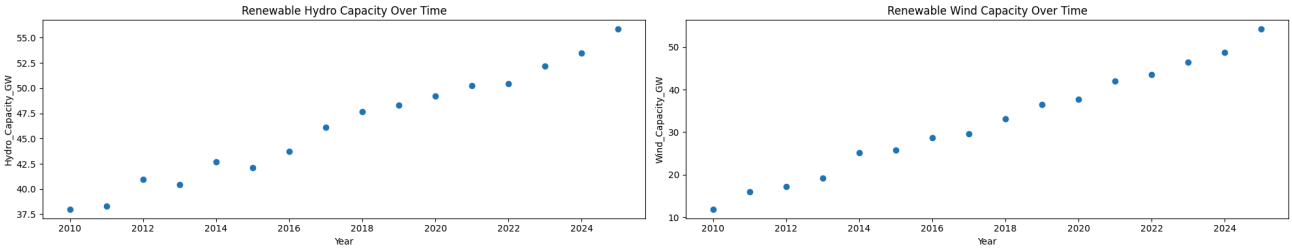
# Renewable Hydro Capacity
axes[0].scatter(
    df_241UDS0096['Year'],
    df_241UDS0096['Hydro_Capacity_GW']
)

axes[0].set_title('Renewable Hydro Capacity Over Time')
axes[0].set_xlabel('Year')
axes[0].set_ylabel('Hydro_Capacity_GW')

# Renewable Wind Capacity
axes[1].scatter(
    df_241UDS0096['Year'],
    df_241UDS0096['Wind_Capacity_GW']
)

axes[1].set_title('Renewable Wind Capacity Over Time')
axes[1].set_xlabel('Year')
axes[1].set_ylabel('Wind_Capacity_GW')

plt.tight_layout()
plt.show()
```



In [ ]:
```

```

Step 3: Comparing Renewable Solar,Wind and Hydro Capacity

In [25]:
```
plt.figure(figsize=(15, 6), dpi=200)

plt.plot(
    df_241UDS0096['Year'],
    df_241UDS0096['Solar_Capacity_GW'],
    label='Solar Capacity'
)

plt.plot(
    df_241UDS0096['Year'],
    df_241UDS0096['Wind_Capacity_GW'],
    label='Wind Capacity'
)

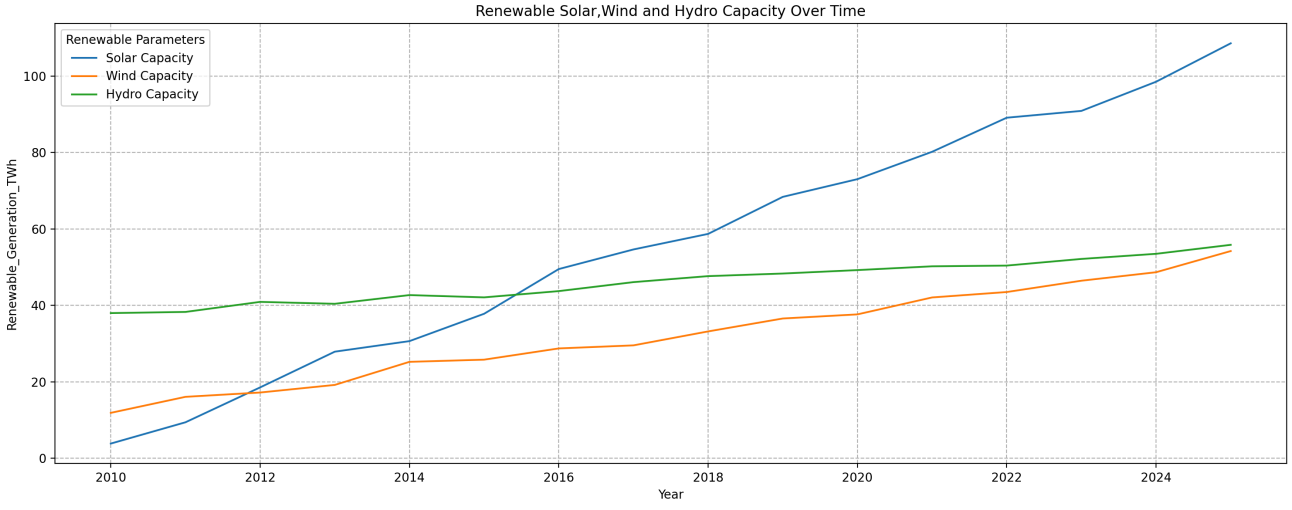
plt.plot(
    df_241UDS0096['Year'],
    df_241UDS0096['Hydro_Capacity_GW'],
    label='Hydro Capacity'
)

plt.title('Renewable Solar,Wind and Hydro Capacity Over Time')
plt.xlabel('Year')
plt.ylabel('Renewable_Generation_TWh')

plt.tight_layout()
plt.legend(
    loc='upper left',
    title='Renewable Parameters'
)

plt.grid(True, linestyle='--')

plt.show()
```



Step 4:InvestmentCrore vs Energy Demand TWh

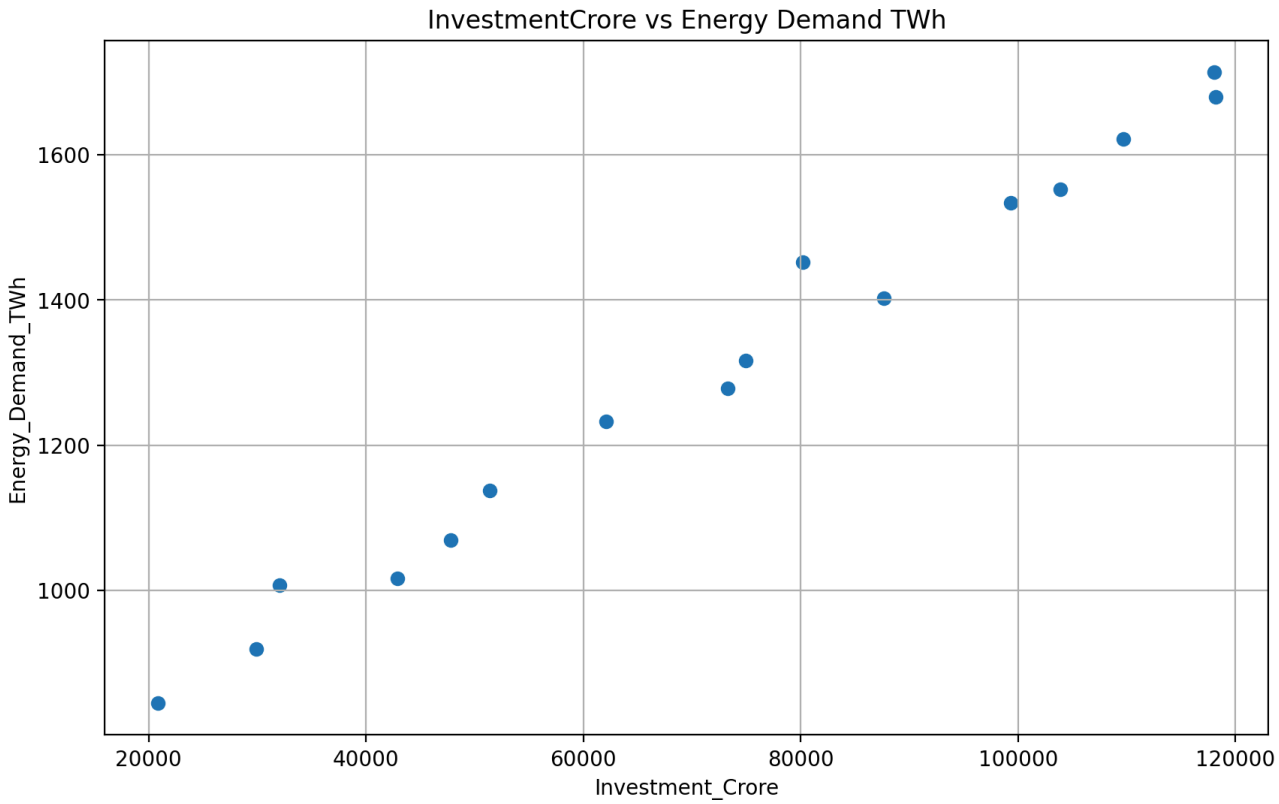
In [22]:
```
plt.figure(figsize=(10, 6), dpi=200)

plt.scatter(
    df_241UDS0096['Investment_Crore'],
    df_241UDS0096['Energy_Demand_TWh']
)

plt.title('InvestmentCrore vs Energy Demand TWh')
plt.xlabel('Investment_Crore')
plt.ylabel('Energy_Demand_TWh')

plt.grid(True)

plt.show()
```



Step 5:Renewable Investment Over Time

In [23]:

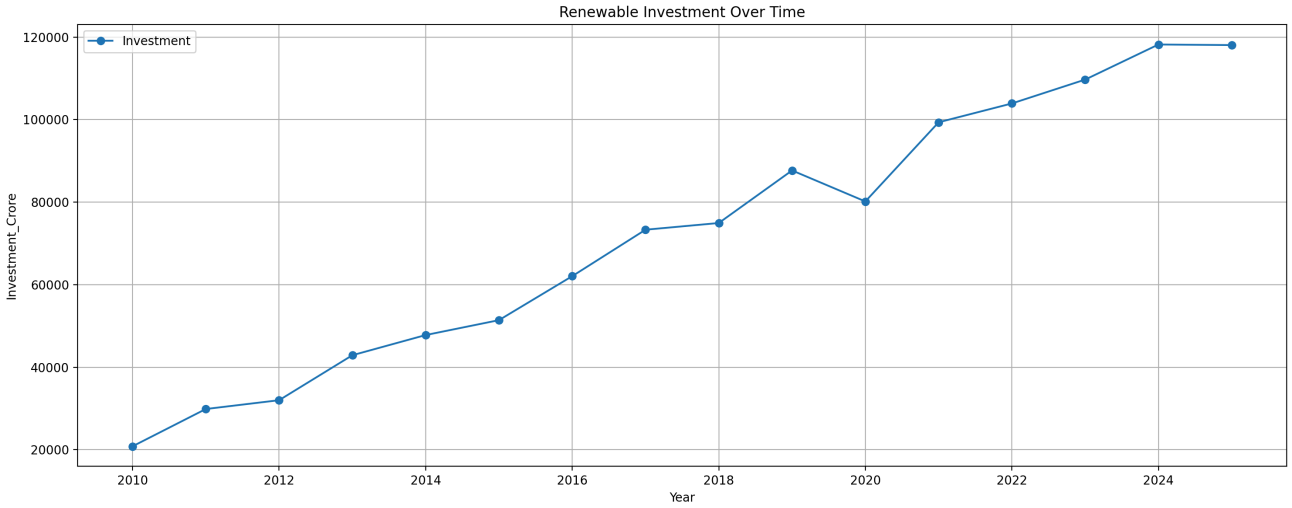
```
plt.figure(figsize=(15, 6), dpi=200)

plt.plot(
    df_241UDS0096['Year'],
    df_241UDS0096['Investment_Crore'],
    marker='o',
    label='Investment'
)

plt.title('Renewable Investment Over Time')
plt.xlabel('Year')
plt.ylabel('Investment_Crore')

plt.tight_layout()
plt.legend()
plt.grid(True)

plt.show()
```



RESULT:

Renewable energy data was successfully analyzed and visualized using Matplotlib.
The analysis shows trends in renewable capacity, generation, energy demand, and investment over the years.

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